

Novel Topological Phase with a Zero Berry Curvature

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We present a two-dimensional (2D) lattice model that exhibits a nontrivial topological phase in the absence of the Berry curvature. Instead, the Berry connection provides the topological nontrivial phase in the model, whose integration over the momentum space, the so-called 2D Zak phase, yields a fractional wave polarization in each direction. These fractional wave polarizations manifest themselves as degenerated edge states with opposite parities in the model.

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Topological insulators (TIs) hold robust edge states when they are put next to vacuum, which renders them promising for applications in electronics and quantum computing [1–3]. So far, three types of TIs are proposed: Chern insulators [4,5], time-reversal-invariant (TRI) TIs [6,7], and topological crystalline insulators [8]. In these systems, the Bloch states generate nonvanishing Berry curvatures (BC), leading to the nontrivial topological indices after momentum integration of the BC [9–12]. The BC is the geometric analogue of the magnetic field in momentum space [13]. However, the nonzero BC is not always a necessary condition for the nontrivial topology of energy bands, if one recalls the situation of an Aharonov-Bohm (AB) effect [14], or a more relevant case of the so-called molecular Aharonov-Bohm (MAB) effect [15], where the electrons are affected by the vector potential in spite of the zero magnetic field (in the AB effect) or zero Berry curvature (in the MAB effect) [16]. The geometric analogue of the vector potential in momentum space corresponds to the Berry connection. Thus, the following question naturally arises: Can a similar situation happen in momentum space that the Berry connection leads to nontrivial topology of energy bands in the absence of BC?

In this Letter, we show nontrivial topological phases in a 2D lattice system in the absence of BC on the basis of the Su-Schrieffer-Heeger (SSH) model [17] with Peierls distortions [18,19]. Here, we characterize the nontrivial topological phases by the integration of the Berry connection over the first Brillouin zone (BZ), which is the 2D Zak phase accompanying the fractional wave polarization in each direction [20]. Different from the TIs such as Chern insulators, TRI TIs and topological crystalline insulators, Berry curvature in this system vanishes everywhere in BZ due to the coexistence of TR and inversion symmetries, except at locations of energy degeneracy, and furthermore

there are no spin or spinlike degrees of freedom in this system. Nontrivial topological edge states associated with this 2D Zak phases are manifested.

The Hamiltonian we consider is given as follows:

$$\hat{\mathcal{H}} = \sum_{i,j} [(t_x + \delta t_x x_{i,j}) c_{i+1,j}^\dagger c_{i,j} + (t_y + \delta t_y y_{i,j}) c_{i,j+1}^\dagger c_{i,j}] + \text{H.c.}, \quad (1)$$

where (i, j) represents a lattice point in a square lattice, $c^\dagger$ and c are the creation and annihilation operators of a spinless electron at the site (i, j) , t_x and t_y are the transfer integral of the equidistant lattice (without distortions) along the x and y direction, respectively, and δt_x and δt_y are the electron-lattice coupling constants. The Peierls distortions are expressed in the following form:

$$x_{i,j} = (-1)^i, \quad y_{i,j} = (-1)^j. \quad (2)$$

According to the parity of site index (i, j) , there are two types of hopping in the x (y) direction such as $t_x - \delta t_x$ ($t_y - \delta t_y$) and $t_x + \delta t_x$ ($t_y + \delta t_y$). One can regard the two types of hopping in each direction as intracellular and intercellular hopping, respectively. Thus there are four atoms in one unit cell as displayed in Fig. 1(a), where the black and red bonds represent the intracellular and intercellular hopping, respectively. For simplicity, we set $t_x = t_y$, $\delta t_x = \delta t_y$ and define $w = t - \delta t$, $v = t + \delta t$. It is evident that our system respects both TR and inversion symmetries.

By applying Fourier transformation for the four atoms in Eq. (1), one obtains a 4×4 matrix $H_{ij}(k)$ in the momentum space, where $H_{12} = H_{34} = w + v \exp(-ik_x)$, $H_{13} = H_{24} = w + v \exp(-ik_y)$, and others are zeros and conjugated

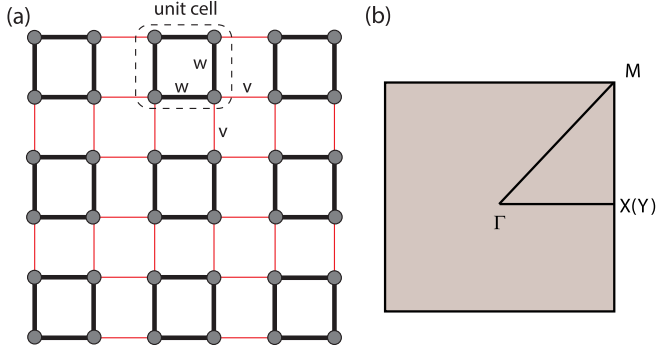


FIG. 1. (a) Schematic of 2D SSH model on a square lattice with symmetric hopping in each direction. Red and black bonds represent the intracell and intercell hoppings, respectively. The amplitudes of intracell and intercell hoppings are w and v , respectively. There are four atoms in one unit cell. (b) First BZ of the square lattice.

components. It is noticed that this Hamiltonian has a zero trace, which is due to the sublattice symmetry. The energy spectrum for $(w, v) = (2.0, 1.0)$ [$(w, v) = (1.0, 2.0)$] is displayed in Fig. 2(a). There are four energy bands, two of them are doubly degenerate at C_{4v} invariant points, and two of them are isolated. Because of the continuous rotational symmetry, each energy band carries an integer angular momentum j (as we consider spinless electrons) and its parity η associated with rotation π is given by $(-1)^j$. Because we consider a square lattice with symmetric hopping, the system also respects C_{4v} point group symmetry (PGS). Thus energy bands with negative parity are doubly degenerate, and energy bands with positive parity are isolated according to the character table of C_{4v} PGS. For conventions, the four energy bands in Fig. 2(a) can be named as s , p_x , p_y , and d_{xy} , which are arranged from bottom to top accordingly. It is noticed that in Fig. 2(a)

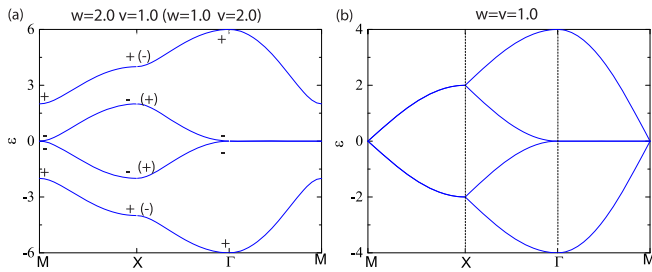


FIG. 2. Bulk spectra of 2D SSH model. From the lowest to the highest the energy bands are s , p_x , p_y , and d_{xy} , accordingly. The two p bands are degenerate at C_{4v} invariant points. Topological phase transition (band inversion) happens when an intracell hopping equals an intercellular one. Bulk spectra before and after the band inversion is shown in (a), where the parities at high symmetric points before and after the band inversion are marked by “ $\pm$ ” with and without parenthesis, respectively. The four bands touch at X (Y) and Γ points when $w = v = 1.0$, which is shown in (b).

degenerate p and s , d_{xy} are separated by pseudolocal gaps [21].

A topological phase transition associated with the 2D Zak phase happens at $|w| = |v|$. Here we show the energy spectrum at the critical point $w = v = 1.0$ in Fig. 2(b). At this critical point, the four energy bands touch at Γ and X (Y) points. The band structure after the band inversion for $w = 1.0$ and $v = 2.0$ is the same as Fig. 2(a) due to the sublattice symmetry. The topology of energy bands is encoded by their parities at high symmetric points, which are marked as “ $\pm$ ” in Fig. 2(a). As displayed in Fig. 2(a), the parity of the lowest s band at X (Y) point changes its sign after the band inversion, which indicates a topological phase transition. This topological phase transition can be characterized by the extended Zak phase in 2D, which is also the wave polarization given by the following expression [22]:

$$\mathbf{P} = \frac{1}{2\pi} \int dk_x dk_y \text{Tr}[\mathbf{A}(k_x, k_y)], \quad (3)$$

where $\mathbf{A} = \langle \psi | i\partial_{\mathbf{k}} | \psi \rangle$ is the Berry connection, and the integration is over the first BZ. Inversion symmetry put a strong constraint on the value of $\mathbf{P}$, which is determined gauge independently by the parities at Γ and X (Y) points as [23]

$$P_i = \frac{1}{2} \left(\sum_n q_i^n \text{modulo } 2 \right), \quad (-1)^{q_i^n} = \frac{\eta(X_i)}{\eta(\Gamma)}, \quad (4)$$

where η denotes the parity, and the summation is over all the occupied bands, and i stands for x or y . Because of C_4 symmetry, we have $P_x = P_y$. After the band inversion one obtains $\mathbf{P} = (1/2, 1/2)$ for the two topological nontrivial band gaps based on Eq. (4).

Different from other TIs [5–8], Berry curvature $\mathcal{F}(\mathbf{k}) = \nabla \times \mathbf{A}$ of Eq. (1) vanishes everywhere in BZ due to the coexistence of TR and inversion symmetries. This is because that the TR symmetry yields $\mathcal{F}(-\mathbf{k}) = -\mathcal{F}(\mathbf{k})$ and the inversion symmetry requires $\mathcal{F}(-\mathbf{k}) = \mathcal{F}(\mathbf{k})$, which gives $-\mathcal{F} = \mathcal{F} = 0$ [13]. For the above argument, the degeneracy of two p bands is not taken into consideration, and they induce singularities in Berry curvature even with a simultaneous presence of TR and inversion symmetries. This energy degeneracy induced singularities of Berry curvature *oscillate fast* along the C_{4v} -invariant line $|k_x| = |k_y|$ [24], which gives a total zero integration over the first Brillouin zone. These singularities play a similar role as a magnetic solenoid in the AB effect or electronic degeneracy in the MAB effect [16]. Besides the vanishing of the Berry curvature of a nondegenerate wave function, one cannot construct two pairs of pseudospins by the combinations of these four orbitals like the topological crystalline insulators [8] as s and d_{xy} bands are not degenerate. Also, the conventional Z_2 number of TRI TIs [25] given by the product of parities at TRI points

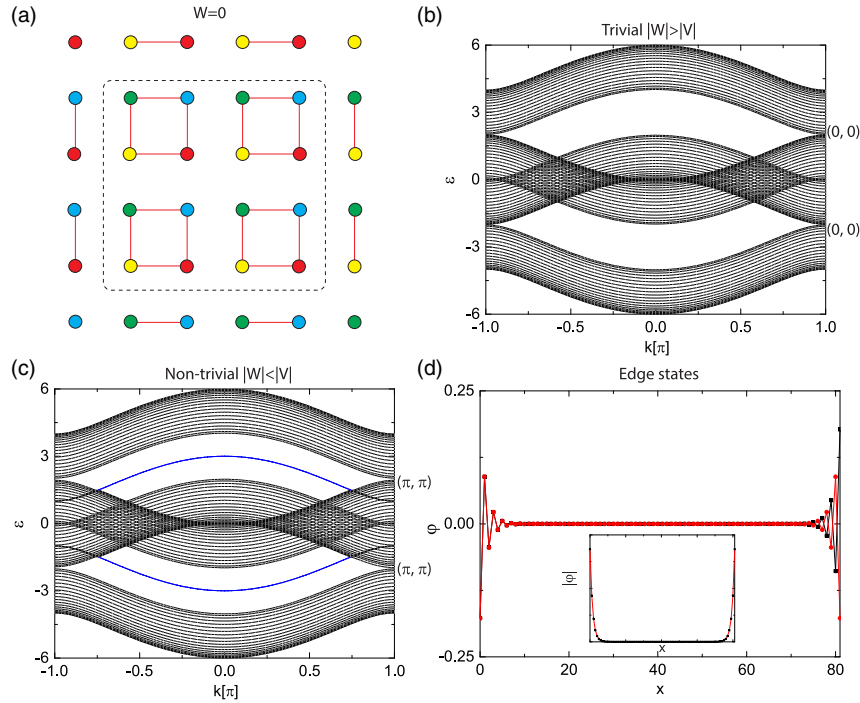


FIG. 3. (a) Edge states naturally appear when the intracellular hopping is 0. There are four types of atoms (marked as different colors in a unit cell) in the edge; thus, there should be four edge states in total. Ribbon spectra for (b) $w = 2.0$, $v = 1.0$, and (c) $w = 1.0$, $v = 2.0$, respectively. After the band inversion, topological edge states marked as blue lines appear in the two topological gaps, which are characterized by the two dimensional Zak phase (π, π) accompanying the fractional wave polarization $(1/2, 1/2)$ for each topological gap as marked by the right side of the panel. The topologically trivial gaps have 2D Zak phase $(0, 0)$. Wave function of edge states at $k = 0.2\pi$ is shown in (d). The edge states are doubly degenerate due to the inversion symmetry along the x direction, which are symmetric and antisymmetric along the x direction. The spatial distribution of the degenerate edge states are shown in the inset of (d).

are always trivial here due to the degeneracy of p modes at Γ and M points. Thus, the system is different from previous 2D TIs, and its topological phase is characterized by the 2D Zak phase (π, π) accompanying the fractional wave polarization $(1/2, 1/2)$. We want to emphasize that the 2D Zak phase or the wave polarization here are totally determined by the bulk property, different from the edge geometry effect studied in Ref. [26], where the 1D Zak phase is determined by the edge shapes.

One essential feature of TIs is the existence of topological edge states when the system is in nontrivial topological phases. This feature is also manifested in our system. Edge states appear naturally when the $|w| < |v|$, for example $w = 0$ as shown in Fig. 3(a). For clarity, we color the four atoms in the same unit cell in Fig. 3(a). It is noticed that there are four types of atoms in the edges; thus, there should be four edge states totally. Figures 3(b) and 3(c) display the ribbon spectra of trivial and nontrivial phases, respectively. Because s and d_{xy} bands are not degenerate, at least two topological gaps are opening simultaneously in our C_{4v} respected system. The edge states are marked as blue lines in Fig. 3(c). It is noticed that the edge states have quadratic dispersions here due to the C_4 PGS, and the ribbon spectra is gapless due to the lacking of complete band gaps in bulk spectrum as displayed in Fig. 1 [27].

From Fig. 3(c) we observe two edge states from the ribbon spectrum in total. However we have four distinct bands in our system, and they are all involved in the topological phase transition. Thus the edge states are doubly degenerate. The wave functions of doubly degenerate edge states are plotted in Fig. 3(d), where one sees that the degenerate edge states are symmetric and antisymmetric along the x direction. This degeneracy is caused by the inversion symmetry along the x direction. The inset of Fig. 3(d) displays the spatial distribution of the degenerate edge states, which coincide with each other. The degeneracy of edge states in this 2D system may yield non-Abelian statistics [28], which remains as a future study topic.

Let us briefly discuss the robustness of the topological edge states. As investigated in the Supplemental Material [29], the difference between two hoppings $|v| - |w|$ determines the robustness of the edge states. For perturbations such as surface potentials and nonuniform on-site energies with amplitudes smaller than $|v| - |w|$, the edge states always exist in the band gaps even for those possessing local band gaps, which is different from the case discussed in Ref. [30]. Furthermore, as the (π, π) phase is protected by the C_{4v} PGS, the edge states can appear in a general direction like the $x = y$ direction. It should be noticed that the band structure of edge states strongly depends on the

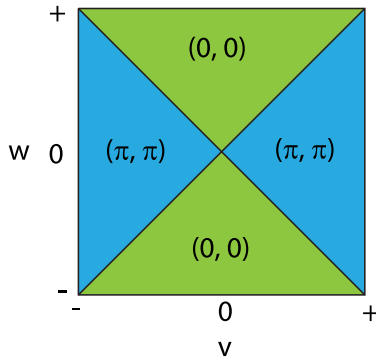


FIG. 4. Topological phase diagram of the 2D SSH model. The system enters topological nontrivial phase when $|w| < |v|$. The nontrivial topological phases are characterized by the non-zero 2D Zak phase (π, π) .

directions of the ribbons. One may also be concerned that the spectrum of edge states shown do not bridge valence and conduction bands, which makes the edge states seemingly trivial. This is because of an open (vacuum) boundary condition used in the calculation, which breaks the C_{4v} PGS. For a nontrivial ribbon bounded by another trivial ribbon possessing C_{4v} symmetry, the edge states bridge the valence and conduction bands.

Finally, we plot the topological phase diagram of our model in Fig. 4. When the intercellular hopping is larger than the intracellular one; the system enters the nontrivial topological phase. The nontrivial/trivial topological phases are characterized by the 2D Zak phase $(\pi, \pi)/(0, 0)$ accompanying fractional wave polarization $(1/2, 1/2)/(0, 0)$.

Although we take the C_{4v} PGS as an example here, the results can be extended to other PGS straightforwardly. For example, the C_{2v} PGS, in which $P_x = P_y$ is not a necessary requirement, and the appearance of edge states will depend on the cutting direction of bulk samples. Besides the square lattice, another case is the hexagonal lattice. For hexagonal case or C_{6v} PGS, there are two pairs of degenerate bands with opposite parities, and a pseudospin-polarized state can be constructed [31]. The topological phases of the hexagonal case can be described by a pseudospin Chern number then. But no matter what kind of lattice is under consideration, the topological phase transition always happens at $|w| = |v|$. It is noticed that we consider electrons here, but the results can be extended to another system such as dielectric photonic crystals [32], which can be more easily be realized by experimentalists due to their macroscopic system sizes.

To summarize, we have shown nontrivial topological phases in the 2D lattice system with zero Berry curvature, which is described by the SSH model in two dimensions. The nontrivial topological phase of the 2D SSH model have been characterized by the 2D Zak phase accompanying the fractional wave polarizations. Bulk edge correspondence associated with the 2D Zak phase is manifested. Our results

will serve to design the new materials with a nontrivial topological phase.

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